

Anand Balachandran

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Projects

Parkinson's Disease Detection Using Voice Biomarkers (Python, scikit-learn, Praat, Streamlit)

- Engineered acoustic biomarkers (pitch, jitter, shimmer, HNR, F0) using Praat and expanded the dataset to 243 samples.
- Built a stacking ensemble (Random Forest, SVM, AdaBoost → XGBoost) achieving 93.88% test accuracy and 0.94 F1-score.
- Developed a Streamlit application for uploading or recording voice samples and performing real-time Parkinson's prediction.
- Built an end-to-end ML pipeline covering feature extraction, training, evaluation, and deployable inference.

Pneumonia Detection Using Deep CNNs (Python, TensorFlow, Keras)

- Trained on 5,800+ chest X-ray images for binary classification (Pneumonia vs Normal).
- Applied transfer learning using VGG16 and ResNet-50 with augmentation and regularization.
- Achieved 95% classification accuracy through hyperparameter tuning and validation.
- Built an end-to-end inference pipeline for real-time medical image prediction.

Advanced Healthcare SQL Analytics (MySQL)

- Designed relational schema and implemented advanced analytical queries using CTEs and window functions.
- Computed revenue segmentation, doctor performance rankings, and monthly growth trends.
- Performed data-driven analysis to identify operational inefficiencies and revenue concentration.
- Analyzed 1,000+ patient and billing records to compute total revenue of 4.93L and identified 1.85L outstanding debt through structured SQL aggregation.

Experience

Frontend Developer Intern, Cropyaan Private Limited

Jan 2026 – Jun 2026

- Built responsive, multi-language (i18n) web pages to expand accessibility and streamline the overall user experience.
- Created a real-time weather advisory feature using the Open-Meteo API to deliver actionable climate insights.
- Designed modular frontend components to improve scalability and maintainability.
- Integrated frontend components with backend APIs and contributed to the development of the Smart Agriculture and Crop Monitoring Platform.

Education

VIT-AP University

Bachelor of Technology in Computer Science Engineering (Core) : 7.62

Aug 2022 – Mar 2026

Amaravati, Andhra Pradesh

Delhi Public School

12th , Central Board of Secondary Education : 80%

Apr 2021 - Jun 2022

Hyderabad, Telangana

Delhi Public School

10th , Central Board of Secondary Education : 88%

Apr 2019 - Mar 2020

Hyderabad, Telangana

Technical Skills

Programming: Python, SQL, Java

Machine Learning & AI: scikit-learn, TensorFlow, Keras, XGBoost, Ensemble Learning, Model Evaluation

Data Processing: Pandas, NumPy, Feature Engineering, Data Cleaning

Visualization: Matplotlib, Tableau

Databases & Tools: MySQL, MongoDB, Git, GitHub

Certifications

- MongoDB Associate Database Administrator
- Oracle Generative AI Professional Certification
- Full Stack Development Training – Blackbucks (2023)